Multimedia Signal Processing - 2nd Module

January 25, 2024

1. **Question (7 pts)**
   Consider two sources X and Y that emit symbols according to the following joint p.m.f.

	y_1	y_2	y_3	y_4
x_1	1/8	0	1/8	0
x_2	0	0	1/4	0
x_3	0	0	1/8	1/4
x_4	0	1/8	0	0

   (a) Compute the probability mass functions for the symbols emitted by the source X.

   (b) Compute the entropy $H(X)$ for the source X.

   (c) Compute the joint entropy $H(X, Y)$ considering both sources X and Y.

   (d) Design the Huffman code to encode single symbols of X and compare the average code length to $H(X)$ and the length of the Shannon code.

   (e) Design the Huffman code to jointly encode symbols of X and Y and compare the average code length to $H(X, Y)$ and the length of the Shannon code.

   (f) How many bits do you need to represent the message $[(x_1, y_1); (x_1, y_3); (x_3, y_3)]$ using the estimated Huffman code?

2. **Question (4 pts)** Consider the problem of video coding.

   (a) Report the block diagram of a typical hybrid DPCM/DCT video codec (i.e., encoder and decoder).

   (b) Describe all the blocks in the diagram.

   (c) Explain when and why motion compensation proves useful compared to a situation in which you do not use it.

3. **Question (4 pts)** Considering the process of defining a codebook for the symbols emitted by a DMS source.

   (a) Define what do we mean by "code".

   (b) Provide the definition of one-to-one, instantaneous, and uniquely decodable code.

   (c) Provide the definition of expected code length.

4. **Questions (8 pts - each answer can be either TRUE or FALSE - Incorrect answers give you -1 points, unanswered 0 points, correct answers 1 point - Provide a brief comment when in doubt)**

   - [T] [F] Considering the same distortion level, transform coding always provides better compression rate than PCM both theoretically and practically.
   - [T] [F] The mutual information $I(X;Y)$ of two sources X and Y must be positive.
   - [T] [F] A closed-loop DPCM is always preferable to a normal PCM scheme.
   - [T] [F] In a video coding scheme, the decoder estimates the motion vectors to reconstruct the frames.
   - [T] [F] Given a discrete source, the average code length of the Shannon code can be equal to the average code length of the Huffman code.
   - [T] [F] Considering a transform coding scheme, KLT transform is always preferable to DCT.
   - [T] [F] A scalar uniform quantizer and a Lloyd-Max quantizer provide the same solution for a uniformly distributed source.
   - [T] [F] In the JPEG standard, the luminance component is often sub-sampled as the chrominance is more important for the human visual system.

5. **Question (Matlab - 10pts)**

   Consider you have an image file called `peppers.png` in your Matlab working directory. This is a RGB image with resolution 512×384.

   Data loading

   (a) Load the image and display it.

   (b) Extract a single color component (of your choice) from the image.

   JPEG coding

   Considering the color component you extracted from the image, implement a simple JPEG encoding and decoding scheme.

   (c) Extract all image non-overlapping blocks of size 8×8.

   (d) Apply 2D DCT to all of them.

   (e) Quantize DCT coefficients with a uniform linear quantizer. Use a quantization step of your choice, justifying the choice.

   (f) Apply inverse DCT to quantized blocks.

   (g) Reconstruct the decoded image and display it.

   (h) Compute the approximate bit-rate (in bit per pixel) that you need to store the image in JPEG using your implementation (hint: consider that you are able to apply an ideal lossless coding scheme in your JPEG implementation).